

Operating Systems and Kernel Design
EECE4811/EECE 5811
Fall 2023

Instructor: Prof. Dalila Megherbi, Room 421B x3307 (Virtual Live via Zoom as scheduled)

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Office Hours: Tuesday: 10:00 AM-noon Thursday: 11:00 AM-noon (Virtual Live via Zoom as scheduled or per Zoom appointment).

Course Description: Covers the components, design, implementation, and internal operations of computer Operating Systems. Topics include the basic structure of operating systems, Kernel, user interface, I/O device management, device drivers, process environment, concurrent processes and synchronization, inter-process communication, process scheduling, memory management, deadlock management and resolution, and file system structures. Laboratories include examples of components design of real operating systems.

Course Prerequisites: EECE, 2160, EECE3220, EECE 3170, or Permission of the Instructor.

Textbook: Operating System Concept, by A. Silberschatz, Latest Edition
Publisher: John Wiley & Sons

Lecture Notes: will be electronically progressively provided.

Grading Policy:	Grad (EECE 5811)	Undergrad (EECE 4811)
Lab Assignment	40%	40%
Exam1	20%	25%
Exam2	30%	33%
Final Project	8%	
<i>Class Attendance:</i>	2%	2%

Tentative Course Outline:

Week#

1. Introduction, Computer-System Structures, Operating-System Structures
2. Processes, Threads; Process/threads Synchronization & Deadlocks
3. Memory management, Virtual Memory
4. CPU Scheduling
5. Operating System Network Structure and its Distributed Communication; Distributed systems and coordination.
6. File Systems
7. Distribution File Systems
8. I/O Systems, Mass-Storage Structure
9. Protection and Security
10. Cloud computing; Big Data Systems/Handling (time allowing)
11. Case Studies (time allowing)
12. Projects

Course mode of delivery: Virtual via Zoom. Currently and unless notified otherwise, the class Zoom URL for students enrolled in the class is

<https://uml.zoom.us/j/2119391870>

- 1- Please connect with your first name and last name as you are enrolled in the class.
- 2- It is best to connect via computer audio; otherwise, enable your microphone.
- 3- By default, you can connect to zoom and call in via phone @+1 312 626 6799,
Participant ID: 49

+1 312 626 6799
+1 646 876 9923
+1 346 248 7799
+1 669 900 6833
+1 253 215 8782
+1 301 715 8592

- 4- Keeping cameras on during class as much as possible and applicable may improve student engagement and help faculty to better assess students' learning. During lectures and discussions, you are expected to keep your cameras on unless you have a valid reason not to, which you should discuss with the Instructor ahead of time. Occasionally turning your camera off during class to attend to a distraction is fine. During exams (if specified by the Instructor to be virtual before the period(s) of examination), cameras are expected to be on all the time throughout the exams.

Policy on Late Assignments: Late assignments will be penalized at 10% off per late submission day if turned in late and will not be accepted after a 5-day delay. However, if there are significant special circumstances, special provisions can be made and if prior arrangements have been made with Prof. Megherbi.

Course Assignments and Exams Schedule: Assignment dates for the whole semester, when they are out and when they are due, as well as exam dates, are provided on a separate hand-out sheet provided in class and posted on the course website.

Academic Integrity: Cheating and plagiarism will not be tolerated. Please review the University policy regarding academic integrity.

Classroom Conduct: Students are expected to exhibit professional and respectful behavior that is conducive to a mutually beneficial learning environment in the classroom. Examples of inappropriate behavior include text messaging, listening to music, cell phone use (other than the campus alert system), late arrivals, early departures, use of laptops for other than class purposes, disrespectful comments or behavior, intentional disruptions, failure to follow faculty directives, etc. Students in violation of these standards may be asked to leave the class.

Class Attendance: Class attendance is mandatory. Attendance is taken randomly in class. Excused absences must be secured in writing (email) in advance from the Instructor.

Technology Resources: For a listing of available computing and software resources available to students, visit: <https://www.uml.edu/IT/Services/DLC/>

Accommodations: Per University policy and the ADA, accommodations are provided for students with documented disabilities. If you have a disability, please contact the Office of Disability Services as soon as possible. Their office is in UC 220 (978-934-4574, Disability@uml.edu). Documentation of disability is confidential. Requests for accommodation

for religious reasons should be directed to Equal Opportunity and Outreach at 978-934-3565, Wannalancit Mills, Suite 301.

University Privacy Statement: UMass Lowell recognizes the importance of mutual trust between students and faculty. Neither faculty nor students may record video or audio of a course or private conversation without all parties consent. Massachusetts is a two-party consent state, which means it is illegal to record someone without their permission. Recordings of classroom lectures, when they occur at the discretion of the Instructor, are the intellectual property of the Instructor. Instructors have the right to prohibit audio and video recording of their lectures. In addition, sharing or selling recordings of classroom activities, discussions, or lectures with any other person or medium without permission of the Instructor is prohibited.

Counseling Services: As part of the Wellness Center, Counseling Services at UMass Lowell provide mental health counseling, consultation, and referrals to help students achieve personal and academic success. They also assist students in better understanding and coping with their feelings, relationships, and choices surrounding their academic success. Visit <https://www.uml.edu/student-services/Counseling/>

University Class Cancellation Information: If the campus is closed (most likely for weather), visit the UML website for relevant announcements.

Diversity, Inclusion, and Classroom Community Standards: your professor values human diversity in all its forms, whether expressed through race and ethnicity, culture, political and social views, religious and spiritual beliefs, language and geographic characteristics, gender, gender identities, and sexual orientations, learning and physical abilities, age, and social or economic classes. Diversity and individual differences are respected, appreciated, and recognized as a source of strength in this class. Please interact respectfully with one another.

University Cancellation Information: If the campus is closed (most likely for weather), visit the website for announcements relevant to the class.

Academic calendar: Please look at the academic calendar for fall 2023 at the UML website <https://www.uml.edu/registrar/calendars/> for the last day (**September 12**) to drop or add a course without a permission number and the last day (**September 19**) to add a course with a permission number directly from the instructor.